drones

MDPI

Article

Energy-Efficient UAV Relaying in Dual Communication Mode

Yuting Hao [1] and Jonathan Kua [2,*]

[1] College of Computer and Information Science, Southwest University, Chongqing 400715, China; hyt2021@email.swu.edu.cn
[2] School of Information Technology, Deakin University, Geelong, VIC 3220, Australia
* Correspondence: jonathan.kua@deakin.edu.au

Highlights

What are the main findings?

- A UAV relay-assisted communication framework with cellular and UAV relay modes is proposed to maximize regional communication capacity by jointly optimizing user connection scheduling, bandwidth allocation, and UAV trajectory under energy and QoS constraints.
- A path-discretization-based alternating algorithm is developed. The UAV trajectory and service time are optimized using DC-SCA, while user connection scheduling and bandwidth allocation are optimized using a penalty-DC method.

What are the implications of the main findings?

- UAV relays can forward ground user data from a heavily loaded cell to a neighboring underutilized BS, thereby improving regional communication capacity.
- Results show that a larger UAV energy budget allows longer communication service and increases regional capacity, while higher QoS requirements place tighter constraints on the UAV relay link.

Abstract

In scenarios with intensive communication demands, heavy ground user (GU) traffic places significant pressure on ground base stations (BSs). Unmanned aerial vehicle (UAV)-assisted cellular communication offers a promising opportunity to enhance regional capacity and quality of service (QoS) by relaying GU data to neighboring underutilized BSs. In this paper, we propose an energy-efficient UAV-assisted relay communication system operating in a dual communication mode (cellular mode and UAV relay mode). Under UAV energy budget constraints, we jointly optimize users' connection scheduling, bandwidth allocation, and UAV trajectory to maximize the long-term throughput of all GUs. The resulting mixed-integer nonlinear non-convex problem is difficult to solve. To address this, we decompose the proposed problem into two subproblems. Specifically, the path discretization technique, successive convex approximation (SCA) method, and difference-of-convex (DC) method are employed to handle trajectory design and variable service durations. The user connection scheduling and bandwidth resource allocation problems are solved via a penalty-DC framework. Simulation results demonstrate that the proposed algorithm achieves higher mean regional communication capacity than the benchmark schemes, while also revealing critical trade-offs between UAV trajectory, onboard energy limits, and user QoS requirements.

Check for updates

Received: 5 August 2026
Revised: 19 October 2026
Accepted: 26 October 2026
Published: 28 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

Keywords: Unmanned Aerial Vehicle (UAV); dual communication modes; path discretization technology

1. Introduction

According to the mobile economy report [1], the total number of Internet users worldwide is expected to reach 6.3 billion by 2030, driven by the rapid development of mobile devices and 5G/6G networks. Concurrently, the surge in connected devices creates severe capacity challenges for traditional terrestrial base stations (BSs). This capacity strain is particularly acute in high-density urban areas, large-scale events, and transportation hubs, where network congestion and service quality degradation are increasingly evident. This problem not only affects users' communication experience but also poses a threat to sustainability [2], making it imperative to quickly and efficiently increase communication network capacity and ensure users' quality of service (QoS). To address these challenges, unmanned aerial vehicles (UAVs) have emerged as a highly flexible solution due to their low cost, controllable mobility, and rapid deployment capabilities [3,4]. By serving as aerial relays, UAVs can forward ground user data from heavily loaded cells to neighboring underutilized BSs. This provides an opportunity to alleviate local network congestion and improve regional communication capacity.

Advances in UAV technology have led to a great deal of research into UAV-enabled communication networks. UAVs serve as multifunctional flight mobile terminals that perform tasks such as surveillance, edge computing, data acquisition, edge caching, and remote sensing [5–13], etc. From a communication perspective, aerial platforms significantly improve wireless links in terms of channel bandwidth, system capacity, and end-user quality of experience (QoE) and QoS [14–18], owing to their favorable line-of-sight (LoS) links, high mobility, and rapid deployment capabilities [19,20].

UAV-assisted communications have been extensively studied over the past decade, but several challenges remain for energy-limited UAV relay platforms:

- **C1:** Energy Constraints on UAV Relay Platforms: UAVs have limited onboard energy, which must support both propulsion and communication. Propulsion energy consumption therefore limits the duration of UAV relay services [21].
- **C2:** Network Architecture Complexity: While UAVs acting as aerial BSs outside terrestrial coverage have been widely studied [22–24], many works overlook the core network backhaul connection or rely on unstable wireless backhaul links in complex environments. Utilizing UAVs as mobile relays to bridge ground users (GUs) with underutilized distant ground BSs provides a connection to the core network through existing BS infrastructure, extending cellular capacity in congested areas [14–18].
- **C3:** Joint Optimization in Dual Communication Modes: Existing UAV relay literature predominantly focuses on simple single-source single-destination pairs [14–18]. Dynamically selecting between cellular mode and UAV relay mode requires sophisticated user connection scheduling and bandwidth allocation under strict QoS constraints.
- **C4:** Coupled Trajectory and Energy Management: The mobility of UAV relays directly alters channel gains and system capacity over time. Efficient trajectory planning while balancing multi-user resource allocation and finite onboard energy budgets presents a challenging non-convex joint design problem.

This paper studies an energy-efficient UAV relay-assisted communication system in dual communication mode, aiming to maximize regional communication capacity while adhering to constraints related to the UAV energy budget and QoS requirements. The focus is the joint design of dual-mode user access and UAV mobility to connect users in a

heavily loaded cell to an underutilized remote BS. The main contributions of this study are summarized below:

- First, this paper establishes a dual-mode communication model in which a heavily loaded BSA serves users jointly with a UAV relay connected to an underutilized BSB. The long-term capacity of communication networks can be maximized by jointly optimizing critical variables: bandwidth allocation, UAV trajectories, and user connection scheduling. At the same time, the model takes into account relevant constraints, including UAV energy limitations and two-hop communication rate constraints. The optimization problem is ultimately a challenging mixed-integer nonlinear non-convex problem.
- Secondly, this paper solves the complex optimization problem by decomposing it into two subproblems. For the UAV trajectory subproblem, path discretization is combined with a difference-of-convex and successive convex approximation (DC-SCA) framework. A penalty-based difference-of-convex (penalty-DC) method is used to solve the user connection scheduling and bandwidth allocation subproblem. To achieve convergence, alternating optimization (AO) techniques are applied to these subproblems. The algorithm uses alternating updates with explicit convergence and scheduling integrality checks to obtain a numerically feasible solution.
- Finally, we conducted extensive simulation experiments and the results show that the proposed algorithm achieves higher mean communication capacity than the benchmark schemes. Furthermore, the results reveal the effects of UAV energy budgets and QoS requirements on trajectory design and communication capacity.

We organize the rest of this paper as follows. Section 2 describes the related work. Section 3 introduces the system model and the problem formulation. Section 4 proposes an efficient solution based on path-discretization-based block coordinate descent (P-BCD), DC-SCA, and penalty-DC optimization techniques. Section 5 presents simulation results and analysis. Finally, Section 6 concludes this paper.

2. Related Works

UAVs have gained attention as a promising technology in enhancing communication connectivity and performing various tasks. The main reasons are its flexible deployment capabilities and low cost.

UAVs serve as versatile flying mobile terminals, undertaking a multitude of tasks including monitoring, edge computing, data collection, edge caching, and remote sensing. Several studies have demonstrated this multifaceted functionality [25,26]. Xu et al. [6] introduced a novel UAV-assisted mobile edge computing (MEC) system aimed at maximizing the weighted computation efficiency. Zeng et al. [8] presented a joint design of layered scalable video caching, user-UAV association, and 3D UAV deployment, with the goal of minimizing cumulative video access delay in cache-enabled multi-UAV networks [27]. Zhan et al. [11] applied the min-max Multiple Traveling Salesman Problem (min-max m-TSP) and convex optimization techniques to optimize data collection completion time in multi-UAV-enabled networks. Samir et al. [13] focused on UAV trajectory planning for efficient data collection from time-constrained IoT devices. These studies collectively highlight the broad range of tasks UAVs can perform. However, they also bring to light a significant limitation: the constrained energy resources of UAVs [28,29]. The engagement of UAVs in additional tasks can substantially reduce their operational service time.

UAVs can also serve as relays or aerial base stations, significantly enhancing wireless communication in terms of capacity, coverage, and reliability. It has been noted that UAVs are particularly well-suited for relay operations due to their ability to carry more portable communication equipment [14–18]. This suitability has led to a surge in research

on UAV relay-assisted communication systems. Zhang et al. [16] explored a half-duplex uplink UAV relay network involving a UAV, a base station (BS), and a mobile device (MD), aiming to minimize the network's outage probability. Zhu et al. [30] focused on maximizing the achievable rate from the source to the destination node in millimeter-wave UAV relay networks, proposing an alternating interference suppression algorithm for optimization with relatively low complexity. Li et al. [31] investigated full-duplex (FD)-based UAV relaying for multiple source-destination pairs, enhancing system capacity and communication coverage. These studies effectively exploit the mobility of drones to create high-quality communication channels between pairs of users. However, these works mainly focus on establishing high-quality channels between individual user pairs, while the aspect of communication from multiple users to a single destination via a UAV relay remains underexplored.

Some research has addressed the communication challenge involving a single source and multiple destinations within UAV relay systems. It is important to note that this scenario can be conceptually equated to a multi-source single-destination problem. The study in [32] introduced a novel scheme to secure a UAV-enabled relaying communication system, which included a base station (BS), multiple users, and an eavesdropper on the ground. Yeom et al. [33] conducted an analysis of the outage probability in a UAV-assisted downlink cellular network comprising a BS and two mobile stations. Additionally, Zhong et al. [34] proposed model-free approaches for optimizing UAV relay deployment to maximize network capacity. Based on the overview of existing research, few works have been dedicated to solving the problem of UAVs and base stations serving users in the same area. Our research addresses this gap and diverges from the aforementioned studies by investigating a joint operational model where UAVs and base stations concurrently cater to users in high-load areas. In this model, GUs have dual communication modes that provide communication performance in densely populated or high-demand areas.

3. System Model and Problem Formulation

In this section, the scenario is first described, followed by the communication and energy models and the optimization problem formulation.

3.1. Scenario Description

As shown in Figure 1, we consider an uplink orthogonal frequency-division multiple access (OFDMA) UAV-assisted cellular network. The system consists of a rotary-wing UAV and two BSs that are connected to the core network. The inclusion of I users in the coverage area of BSA results in a high load. The set of ground users is denoted by $\mathcal{I} = \{1, \ldots, I\}$.

The GU_i^A is defined as the i-th ground user in the coverage area of BSA. Correspondingly, BSB is idle due to the small number of connected users. We assume that the two BSs are at a given altitude $H_{bs} > 0$ and the ground users are at zero altitude. $\mathbf{q}_A = [x_A, y_A]^T \in \mathbb{R}^{2\times 1}$ and $\mathbf{q}_B = [x_B, y_B]^T \in \mathbb{R}^{2\times 1}$ represent the horizontal locations of the two BSs, with $[\cdot]^T$ denoting the transpose, respectively. Similarly, GU_i^A has a horizontal coordinate denoted by $\mathbf{q}_i^A = [x_i^A, y_i^A]^T \in \mathbb{R}^{2\times 1}$. To improve communication capacity in the high-load area, we introduce one UAV as an ideal two-hop data relay. It is assumed that the UAV flies at a fixed altitude $H_{uav} > 0$ to avoid frequent descent and ascent, which corresponds to the minimum altitude that is suitable for the terrain [35]. The total UAV service time is denoted as T, and the UAV horizontal coordinate is denoted by $\mathbf{q}(t) = [x_u(t), y_u(t)]^T \in \mathbb{R}^{2\times 1}$ at any time instant $t \in [0, T]$. As such, the velocity and acceleration vectors of the UAV are represented by $v(t) \triangleq \dot{\mathbf{q}}(t)$ and $a(t) \triangleq \ddot{\mathbf{q}}(t)$, respectively. We consider the two communication modes in the network.

- **Cellular mode:** The GU_i^A communicates with the BSA directly.

- **UAV relay mode:** The GU_i^A transmits data to the UAV relay, which forwards the data to the remote idle BSB.

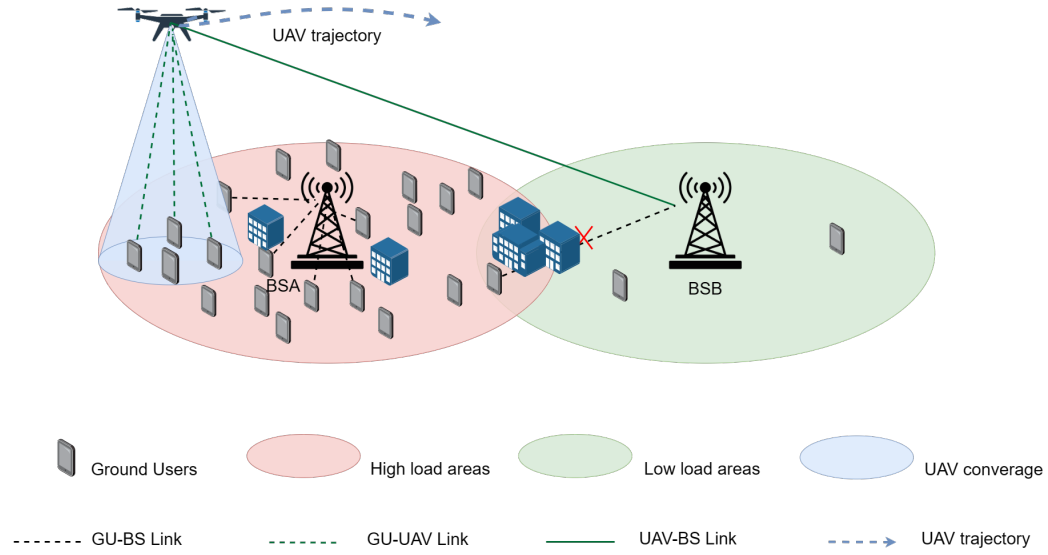


Figure 1. Illustration of the dual-mode UAV relay-assisted cellular network.

Here, we introduce a binary variable $x_{ia}(t) \in \{0,1\}$, which indicates that GU_i^A selects the cellular mode if $x_{ia}(t) = 1$; otherwise, $x_{ia}(t) = 0$. Similarly, $x_{iu}(t) \in \{0,1\}$ is also a binary variable, expressing that the GU_i^A communication mode is the UAV relay mode. It is assumed that the GU_i^A can only select one of the communication modes or not connect at all, which means $x_{ia}(t) + x_{iu}(t) \leq 1, \forall i$. Furthermore, the numbers of users connected to the UAV and BSA are limited. We assume that the maximum numbers of connections for BSA and the UAV are x_{max}^a and x_{max}^u, respectively. Therefore, there are $\sum_{i=1}^{I} x_{ia}(t) \leq x_{max}^a$ and $\sum_{i=1}^{I} x_{iu}(t) \leq x_{max}^u$. Since the OFDMA protocol is used between GU_i^A and either BSA or the UAV, we assume $\alpha_i(t) \in [0,1]$ denotes the fraction of bandwidth assigned to GU_i^A. Note that the sum of bandwidth allocation ratios cannot exceed 1 at time t, and thus we have $\sum_{i=1}^{I} \alpha_i(t) \leq 1$.

3.2. Communication Model

(1) Cellular mode: We consider the uplink where the GU_i^A connects directly to the base station by OFDMA. It is worth noting that, in reality, communication links are often obscured by buildings, plants, etc., so we consider both LoS and non-line-of-sight (NLoS) links. Thus, the channel gain between GU_i^A and BSA at time instant t is expressed as $h_{ia}(t) = \sqrt{\beta_{ia}(t)} \cdot \widetilde{h}_{ia}(t)$, where $\widetilde{h}_{ia}(t)$ is the small-scale fading [36]. The small-scale fading coefficient $\widetilde{h}_{ia}(t)$ is modeled using Rician fading satisfying $\mathbb{E}\{|\widetilde{h}_{ia}(t)|^2\} = 1$. $\beta_{ia}(t)$ represents the large-scale channel coefficient, which is described as

$$\beta_{ia}(t) = \begin{cases} \beta_0 d_{ia}^{-\widetilde{\alpha}}(t) & \texttt{LoS link}, \\ \kappa \beta_0 d_{ia}^{-\widetilde{\alpha}}(t) & \texttt{NLoS link}, \end{cases} \tag{1}$$

where β_0 is the channel power gain at a reference distance of $d_0 = 1$ m, $d_{ia}(t)$ denotes the distance between GU_i^A and BSA, which is defined as $d_{ia}(t) = \sqrt{H_{bs}^2 + \|\mathbf{q}_A - \mathbf{q}_i^A\|^2}$, $\widetilde{\alpha}$ is the path loss exponent and $\kappa < 1$ is an additional attenuation factor due to the NLoS link. Let $P_{\text{LoS}}(t)$ denote the LoS probability between GU_i^A and BSA at time t, with $P_{\text{LoS}}(t) = \frac{1}{1+C \cdot \exp(-D[\theta - C])}$, where C and D are the parameters relating to the environmental condition [35], and $\theta(t) = \arcsin(\frac{H_{bs}}{d_{ia}(t)})$ is the elevation angle at time instant

t. As a result, the expected channel gain is given by $\mathbb{E}\{|h_{ia}(t)|^2\} = \hat{P}_{\text{LoS}}(t)\beta_0 d_{ia}^{-\tilde{\alpha}}(t)$, where $\hat{P}_{\text{LoS}}(t) = P_{\text{LoS}}(t) + (1 - P_{\text{LoS}}(t))\kappa$ can be regarded as a regularized LoS probability. Based on the above models, let $P_i(t)$ and B denote the transmission power of GU_i^A and the total system bandwidth in hertz (Hz). Thus, the achievable rate from GU_i^A to the BSA is defined as $\hat{R}_{ia}(t) = \alpha_i(t)B\log_2\left(1 + \frac{P_i(t)|h_{ia}(t)|^2}{N_0\alpha_i(t)B}\right)$, where N_0 is the noise power spectral density in W/Hz, and $N_0\alpha_i(t)B$ is the noise power over the allocated bandwidth. The $\hat{R}_{ia}(t)$ is also a random variable because of the random channel gain $h_{ia}(t)$. Therefore, we pay attention to the expected communication rate, which is defined as $\mathbb{E}\{\hat{R}_{ia}(t)\}$. Using Jensen's inequality, we have $\mathbb{E}\{\hat{R}_{ia}(t)\} \leq \alpha_i(t)B\log_2\left(1 + \frac{P_i(t)\mathbb{E}\{|h_{ia}(t)|^2\}}{N_0\alpha_i(t)B}\right) = \alpha_i(t)B\log_2\left(1 + \frac{P_i(t)\tilde{\gamma}_0\hat{P}_{\text{LoS}}(t)}{\alpha_i(t)(H_{bs}^2 + \|\mathbf{q}_A - \mathbf{q}_i^A\|^2)^{\alpha}}\right)$, where $\tilde{\gamma}_0 \triangleq \frac{\beta_0}{N_0 B}$ and $\alpha \triangleq \frac{\tilde{\alpha}}{2}$. Note that the $\hat{P}_{\text{LoS}}(t)$ is a complicated function on account of the elevation angle. Thus, we use the homogeneous approximation of $\hat{P}_{\text{LoS}}(t)$, i.e., $\hat{P}_{\text{LoS}}(t) \simeq \bar{P}_{\text{LoS}}(t), t \in [0, T]$, where $\bar{P}_{\text{LoS}}(t)$ is the value corresponding to the most likely elevation angle [37]. As a result, the approximation of the expected communication rate from GU_i^A to BSA can be written as

$$R_{ia}(t) = \alpha_i(t)B\log_2\left(1 + \frac{P_i(t)\gamma_0}{\alpha_i(t)(H_{bs}^2 + \|\mathbf{q}_A - \mathbf{q}_i^A\|^2)^{\alpha}}\right), \tag{2}$$

where $\gamma_0 \triangleq \tilde{\gamma}_0\bar{P}_{\text{LoS}}(t)$.

(2) UAV relay mode: For a GU_i^A selecting the UAV relay mode, there are two links: the GU_i^A to the UAV relay and the UAV relay to the remote idle BSB. It is worth noting that the communication channel is dominated by the LoS link when the UAV is at a medium altitude [38–40]. To simplify the model, we assume that the channel model conforms to a distance-dependent path loss model and that the Doppler effect due to the mobility of the UAV is fully compensated at the receiver. Therefore, the LoS channel gain from the GU_i^A to the UAV can be expressed as $h_{iu}(t) = \beta_0 d_{iu}^{-\tilde{\alpha}}(t)$, where $d_{iu}(t) = \sqrt{H_{uav}^2 + \|q_i^A - q(t)\|^2}$ is the distance between the GU_i^A and the UAV. Correspondingly, the instantaneous achievable rate from GU_i^A to the UAV with OFDMA is given by

$$R_{iu}(t) = \alpha_i(t)B\log_2\left(1 + \frac{P_i(t)\gamma_0^{'}}{\alpha_i(t)(H_{uav}^2 + \|\mathbf{q}_i^A - \mathbf{q}(t)\|^2)^{\alpha}}\right), \tag{3}$$

where $\gamma_0^{'} \triangleq \frac{\beta_0}{BN_0} = \tilde{\gamma}_0$.

Similarly, let $P_u(t)$ denote the transmission power of the UAV, and the instantaneous achievable rate from the UAV to the BSB can be expressed as

$$R_{ub}(t) = B\log_2\left(1 + \frac{P_u(t)\beta_0}{N_0 B(\|H_{uav} - H_{bs}\|^2 + \|\mathbf{q}(t) - \mathbf{q}_B\|^2)^{\alpha}}\right). \tag{4}$$

We assume that the UAV relay is equipped with a large enough data buffer and forwards the received data to the remote idle BSB. Since it takes two hops for GU_i^A to connect to BSB via the UAV relay, the communication rate that can be improved by the UAV relay in high-load areas has the following constraints:

$$R_u(t) = \min\{\sum_{i=1}^{I} x_{iu}(t)R_{iu}(t), R_{ub}(t)\}. \tag{5}$$

3.3. Energy Consumption Model

In this paper, we consider the total UAV energy consumption, which includes the UAV propulsion energy consumption and communication energy consumption. For a rotary-wing UAV, the propulsion energy consumption can be described as [41]

$$E_{\mathrm{p}} \approx \int_0^T \left(P_0 + \frac{3P_0\|v\|^2}{U_{tip}^2} + \frac{1}{2} d_0 \rho s A \|v\|^3 \right) dt + \int_0^T P_i \left(\sqrt{1 + \frac{\|v\|^4}{4v_0^4}} - \frac{\|v\|^2}{2v_0^2} \right)^{\frac{1}{2}} dt, \tag{6}$$

where P_0 and P_i denote the blade profile power and induced power in hover, respectively. v, v_0 and U_{tip} are the UAV flight speed, the average rotor-induced velocity in hover and the tip velocity of the rotor blades, respectively. Furthermore, d_0 and ρ denote the fuselage drag ratio and air density, respectively. Additionally, s and A are defined as the rotor solidity and the rotor disc area, respectively. Therefore, the total UAV energy consumption can be expressed as $E_{\text{total}} = E_{\mathrm{p}} + \int_{t=0}^{T} P_u(t)dt$.

The main mathematical notations used in the system model and optimization formulation are summarized in Table 1.

Table 1. Summary of main notations and definitions.

Notation	Definition
I	The number of ground users
T	The UAV service time
t	Time instant
$\mathbf{q}(t)$	The UAV horizontal coordinate at time instant t
$x_{ia}(t)$	Binary variable indicating that the GU_i^A selects the cellular mode at time instant t
$x_{iu}(t)$	Binary variable indicating that the GU_i^A selects the UAV relay mode at time instant t
$\alpha_i(t)$	The fraction of bandwidth assigned to the GU_i^A at time instant t
$R_{ia}(t)$	The expected communication rate from the GU_i^A to BSA at time instant t
$R_{iu}(t)$	The instantaneous achievable rate from the GU_i^A to the UAV at time instant t
$R_{ub}(t)$	The instantaneous achievable rate from the UAV to the BSB at time instant t
$R_u(t)$	The maximum instantaneous achievable rate from the scheduled GUs to the BSB at time instant t
E_{p}	The propulsion energy consumption of the UAV (J)
B	The total bandwidth (Hz)
$v(t)$	The UAV velocity at time instant t (m/s)
$a(t)$	The UAV acceleration at time instant t ($\mathrm{m/s^2}$)
$P_{\mathrm{LoS}}(t)$	The probability of LoS links between the GU_i^A and BSA at time instant t
$\mathbf{q}_n$	The UAV coordinate at the n-th waypoint
τ_n	The time duration along the n-th line segment

3.4. Problem Formulation

In order to maximize the regional communication capacity, our goal is to maximize system capacity under the QoS and the energy constraints as well as two-hop rate constraints. In this paper, we jointly optimize the UAV service time $\mathbf{T}$, the bandwidth allocation factor $\mathbf{A} = \{\alpha_i(t), \forall i, t\}$, the user connection scheduling $\mathbf{X} = \{x_{ia}(t), x_{iu}(t), \forall i, t\}$, and the trajectory of the UAV $\mathbf{Q} = \{q(t), \forall i, t\}$. The optimization problem is formulated as

$$(\mathrm{P1}): \max_{\mathbf{T},\mathbf{A},\mathbf{X},\mathbf{Q}} \int_0^T \sum_{i=1}^{I} x_{ia}(t) R_{ia}(t) dt + \int_0^T R_u(t) dt \tag{7}$$

$$\text{s.t.} \sum_{i=1}^{I} \alpha_i(t) \leq 1, \alpha_i(t) \in [0,1], \forall i, t \in [0,T], \tag{8}$$

$$x_{ia}(t), x_{iu}(t) \in \{0,1\}, \forall i, t \in [0,T], \tag{9}$$

$$\sum_{i=1}^{I} x_{ia}(t) \leq x_{max}^{a}, \forall t \in [0,T], \tag{10}$$

$$\sum_{i=1}^{I} x_{iu}(t) \leq x_{max}^{u}, \forall t \in [0,T], \tag{11}$$

$$x_{ia}(t) + x_{iu}(t) \leq 1, \forall i, t \in [0,T], \tag{12}$$

$$\frac{\sum_{i=1}^{I} x_{ia}(t) R_{ia}(t)}{x_{max}^{a}} \geq R_{min}, \forall t \in [0,T], \tag{13}$$

$$\frac{R_u(t)}{x_{max}^{u}} \geq R_{min}, \quad \forall t \in [0,T]. \tag{14}$$

$$E_{\text{total}} \leq E_{\max}, \tag{15}$$

$$\|v(t)\| \leq v_{max}, \quad \|a(t)\| \leq a_{max}, \quad q(0) = q(T). \tag{16}$$

where $E_{\max}$ and v_{max} are the maximum usable energy and velocity, respectively. The maximum acceleration is a_{max}. Constraints (9)–(12) are the user connection scheduling constraints, and constraint (12) indicates that the GU_i^A can only select one communication mode or none of them. Constraints (13) and (14) specify the minimum average rate per available connection for the cellular and UAV relay modes, respectively. Constraint (15) corresponds to the total UAV energy constraint, and constraint (16) describes the return of the UAV to its initial position at the end of time, as well as speed and acceleration constraints. Note that problem (7) is a mixed-integer non-convex problem that is hard to solve for the following reasons. First, the optimization problem involves an unknown upper limit variable $\mathbf{T}$ and infinitely many time-dependent optimization variables $\mathbf{Q}$. Second, it contains two binary variables $x_{ia}(t), x_{iu}(t)$, as well as multiple non-convex constraints. Third, UAV trajectory and communication-related variables need to be jointly optimized. Based on the above reasons, it is difficult to get an optimal solution to the problem (7). So, in the following section, we propose an efficient algorithm to obtain suboptimal solutions.

4. Proposed Approach

In this section, we notice that UAV trajectory optimization variables and communication-related optimization variables are strongly coupled. In order to solve the above problem, we first use the block coordinate descent (BCD) method to decouple related variables. Specifically, the first subproblem uses path discretization to handle the unknown integral upper bound $\mathbf{T}$ and the time-dependent UAV trajectory variables $\mathbf{Q}$. The second subproblem optimizes bandwidth allocation and user connection scheduling for a given UAV

trajectory **Q** and service time **T**. We then introduce an alternating optimization algorithm to alternately optimize the two subproblems until convergence.

4.1. UAV Trajectory and Service Time Optimization

With fixed user connection scheduling $\mathcal{X}$ and bandwidth allocation $\mathcal{A}$, problem (7) can be written as

$$(\text{P2}): \max_{\mathbf{Q},\mathbf{T}} \int_0^T \sum_{i=1}^{I} x_{ia}(t) R_{ia}(t) dt + \int_0^T R_u(t) dt \tag{17}$$

$$\text{s.t.} \quad (14) - (16). \tag{18}$$

Problem (17) is intractable due to infinite time-varying optimization variables and an unknown upper limit of integration. To solve this problem, we use path discretization technology, which does not require predetermining the total service time of the UAV compared with traditional time discretization technology [35,42,43]. The path discretization technique is to discretize the UAV trajectory into a limited number of line segments, where each line segment has a variable duration. Specifically, we discretize the UAV path into N line segments, where there are $N+1$ waypoints $\{q_n, 0 \le n \le N\}$, and the time duration of each line segment is expressed as $\{\tau_n, 1 \le n \le N\}$. Therefore, the UAV trajectory can be represented by the waypoints $\{q_n\}$ and the time duration $\{\tau_n\}$, and the optimization variable T can be represented by $T = \sum_{n=1}^{N} \tau_n$. We assume that $\|q_n - q_{n-1}\| \le d_{max}, 1 \le n \le N$, and when d_{max} is small enough, the location and velocity of the UAV in each line segment can be approximately unchanged, so we have $v_n = \frac{q_n - q_{n-1}}{\tau_n}, 1 \le n \le N$. Accordingly, the UAV's total energy consumption can be approximated as

$$\begin{aligned} \widetilde{E}_{total} = & \sum_{n=1}^{N} \tau_n \left(P_0 + \frac{3P_0 \|v_n\|^2}{U_{tip}^2} + \frac{1}{2} d_0 \rho s A \|v_n\|^3 \right) \\ & + \sum_{n=1}^{N} \tau_n P_i \left(\sqrt{1 + \frac{\|v_n\|^4}{4v_0^4}} - \frac{\|v_n\|^2}{2v_0^2} \right)^{\frac{1}{2}} + \sum_{n=1}^{N} \tau_n P_{un}. \end{aligned} \tag{19}$$

We assume that P_{in}, P_{un} and α_{in} are the discrete forms corresponding to $P_i(t)$, $P_u(t)$ and $\alpha_i(t)$ respectively. Accordingly, the discrete form of $R_{ia}(t)$ along the nth line segment is R_{ian}, i.e., $R_{ian} = \alpha_{in} B \log_2 \left(1 + \frac{P_{in} \gamma_0}{\alpha_{in} (H_{bs}^2 + \|\mathbf{q}_A - \mathbf{q}_i^A\|^2)^{\alpha}} \right)$. Correspondingly, we can obtain other discrete forms R_{iun}, R_{ubn} and R_{un}. Note that $R_{un} = min\{\sum_{i=1}^{I} x_{iun} R_{iun}, R_{ubn}\}$ is a nonlinear function. We can transform it into the following two inequality constraints

$$R_{un} \le \sum_{i=1}^{I} x_{iun} R_{iun}, \forall n, \tag{20}$$

$$R_{un} \le R_{ubn}, \forall n, \tag{21}$$

Let $\mathcal{Q} = \{q_n, 0 \le n \le N\}$, $\mathcal{V} = \{v_n, 1 \le n \le N\}$ and $\mathcal{T} = \{\tau_n, 1 \le n \le N\}$. With the path discretization technology, the optimization problem (17) can be approximated as

$$(\text{P3}): \max_{\mathcal{Q},\mathcal{V},\mathcal{T},\{R_{un}\}} \sum_{i=1}^{I} \sum_{n=1}^{N} \tau_n x_{ian} R_{ian} + \sum_{n=1}^{N} \tau_n R_{un} \tag{22}$$

$$\text{s.t.} \quad \frac{R_{un}}{x_{max}^u} \ge R_{min}, \quad \forall n. \tag{23}$$

$$\widetilde{E}_{total} \le E_{\max}, \tag{24}$$

$$\begin{aligned} q_n - q_{n-1} &= \tau_n v_n, \quad \tau_n > 0, \quad \|v_n\| \le v_{max}, \quad n = 1, \ldots, N, \\ 2\|v_n - v_{n-1}\| &\le a_{max}(\tau_n + \tau_{n-1}), \quad n = 2, \ldots, N, \\ 2\|v_1 - v_N\| &\le a_{max}(\tau_1 + \tau_N). \end{aligned} \tag{25}$$

$$\|q_n - q_{n-1}\| \le d_{max}, \quad q_0 = q_N. \tag{26}$$

$$\text{s.t.} \quad (20) \text{ and } (21). \tag{27}$$

Intuitively, the length of each line segment does not exceed a certain threshold, which can ensure that the problem (22) using path discretization technology has sufficiently high finite-sum approximation accuracy. Lemma 1 describes the finite-sum approximation error.

Lemma 1. *For each segment n, hold the scheduling, bandwidth allocation, and transmission powers fixed, and let D_{qn} be a finite Lipschitz constant of the physical relay rate $R_u(q)$ along that segment. Such a constant can be chosen as the larger of the gradient-norm bounds of the aggregate access rate and the backhaul rate; their minimum remains Lipschitz even where it is not differentiable. Let $D_q = \max_n D_{qn}$ and $d_{max} \ge \max_n \|q_n - q_{n-1}\|$. In the following approximation bound, $R_{un}(q_n)$ denotes the physical relay rate evaluated at the segment endpoint, rather than an auxiliary rate below that value. Given T and the maximum distance d_{max} between consecutive waypoints, $\int_0^T R_u(q(t))dt$ can be approximately expressed as $\sum_{n=1}^N \tau_n R_{un}(q_n)$. The upper bound on the finite-sum approximation error is given by*

$$e = |\int_0^T R_u(q(t))dt - \sum_{n=1}^{N} \tau_n R_{un}(q_n)| \le \frac{1}{2} D_q d_{max} T. \tag{28}$$

Proof. See Appendix A. □

The non-concave objective function and multiple non-convex constraints make problem (22) difficult to solve. To solve this problem, we use the BCD technique to divide it into two subproblems. In the first subproblem, we use SCA to optimize the UAV waypoints $\mathcal{Q}$ and UAV's speed $\mathcal{V}$. The second subproblem uses DC-SCA technology to optimize UAV waypoint $\mathcal{Q}$ and time duration $\mathcal{T}$.

4.1.1. UAV Waypoint and Speed Optimization

With fixed user connection scheduling $\mathcal{X}$, bandwidth allocation $\mathcal{A}$, and time durations $\mathcal{T}$, problem (22) can be written as

$$(\text{P3.1}): \max_{\mathcal{Q},\mathcal{V},\{R_{un}\}} \sum_{i=1}^{I} \sum_{n=1}^{N} \tau_n x_{ian} R_{ian} + \sum_{n=1}^{N} \tau_n R_{un} \tag{29}$$

$$\text{s.t.} \quad (20) \text{ and } (21), \ (23)\text{–}(26). \tag{30}$$

where

$$R_{iun} = \alpha_{in} B \log_2 \left(1 + \frac{P_{in}\gamma_0'}{\alpha_{in}(H_{uav}^2 + \|\mathbf{q}_n - \mathbf{q}_i^A\|^2)^\alpha}\right), \forall i, n, \tag{31}$$

$$R_{ubn} = B \log_2 \left(1 + \frac{P_{un}\beta_0}{N_0 B((H_{uav} - H_{bs})^2 + \|\mathbf{q}_n - \mathbf{q}_B\|^2)^\alpha}\right), \forall n. \tag{32}$$

Problem (29) is non-convex due to the non-concave objective function and non-convex constraints (20) and (21), and (23) and (24). We note that the non-convexity of the constraints (20) and (21) is due to the UAV trajectory q_n. However, in particular, R_{iun} and R_{ubn} are convex functions with respect to $\|\mathrm{q}_n - \mathrm{q}_i^A\|^2$ and $\|\mathrm{q}_n - \mathrm{q}_B\|^2$ respectively. So, we can use SCA technology to get the global lower bound by their first-order Taylor expansion at any

point. At the r-th iteration, a lower bound on R_{iun} can be obtained at the given local point q_n^r, formulated as

$$\begin{aligned} R_{iun} &= \alpha_{in} B \log_2\left(1 + \frac{P_{in}\gamma_0'}{\alpha_{in}(H_{uav}^2 + \|\mathbf{q}_n - \mathbf{q}_i^A\|^2)^{\alpha}}\right) \\ &\geq -\mathcal{A}_{iun}^r(\|\mathbf{q}_n - \mathbf{q}_i^A\|^2 - \|\mathbf{q}_n^r - \mathbf{q}_i^A\|^2) + \mathcal{B}_{iun}^r \\ &\triangleq \mathcal{R}_{iun}^{\mathrm{lb,r}}, \end{aligned} \tag{33}$$

$\mathcal{A}_{iun}^r = \frac{B\alpha_{in}P_{in}\gamma_0'\alpha\log_2 e}{\Omega(\alpha_{in}\Omega^{\alpha}+P_{in}\gamma_0')}$ and $\mathcal{B}_{iun}^r = \alpha_{in}B\log_2\left(1+\frac{P_{in}\gamma_0'}{\alpha_{in}\Omega^{\alpha}}\right)$ in (33), where $\Omega \triangleq H_{uav}^2 + \|q_n^r - q_i^A\|^2$ and $\mathcal{R}_{iun}^{\mathrm{lb,r}}$ is the lower bound. Likewise, by replacing the corresponding parameters in R_{ubn}, we can also derive the lower bound $\mathcal{R}_{ubn}^{\mathrm{lb,r}}$ at the given local point q_n^r. Furthermore, note that (24) is a non-convex constraint and the non-convexity is due to $\left(\sqrt{1+\frac{\|v_n\|^4}{4v_0^4}} - \frac{\|v_n\|^2}{2v_0^2}\right)^{\frac{1}{2}}$. Thus, to handle the non-convexity, we introduce slack variables $\{o_n > 0\}$ such that $o_n = \left(\sqrt{1+\frac{\|v_n\|^4}{4v_0^4}} - \frac{\|v_n\|^2}{2v_0^2}\right)^{\frac{1}{2}}$, which is equivalent to $\frac{1}{o_n^2} = o_n^2 + \frac{\|v_n\|^2}{v_0^2}$. We define $\widetilde{E}_{\mathrm{p}}(v_n, o_n) \triangleq \sum_{n=1}^{N}\tau_n\left(P_0 + \frac{3P_0\|v_n\|^2}{U_{tip}^2} + \frac{1}{2}d_0\rho sA\|v_n\|^3 + P_i o_n\right)$ and introduce the following constraint:

$$o_n^2 + \frac{\|v_n\|^2}{v_0^2} \geq \frac{1}{o_n^2}. \tag{34}$$

Since $\|v_n\|^2$ and o_n^2 are convex functions of v_n and o_n, respectively, their first-order Taylor expansions at the feasible reference point (v_n^r, o_n^r) give

$$\begin{aligned} o_n^2 + \frac{\|v_n\|^2}{v_0^2} &\geq (o_n^r)^2 + 2o_n^r(o_n - o_n^r) \\ + \frac{\|v_n^r\|^2}{v_0^2} + \frac{2}{v_0^2}(v_n^r)^T(v_n - v_n^r) &\triangleq \Lambda^{lb,r}. \end{aligned} \tag{35}$$

Based on the above discussion, problem (22) can be approximately expressed as

$$(\mathrm{P4}): \max_{\mathcal{Q},\mathcal{V},\{R_{un}\},\{o_n\}} \sum_{i=1}^{I}\sum_{n=1}^{N}\tau_n x_{ian}R_{ian} + \sum_{n=1}^{N}\tau_n R_{un} \tag{36}$$

$$\mathrm{s.t.} R_{un} \leq \sum_{i=1}^{I} x_{iun}\mathcal{R}_{iun}^{\mathrm{lb,r}}, \forall n, \tag{37}$$

$$R_{un} \leq \mathcal{R}_{ubn}^{\mathrm{lb,r}}, \forall n, \tag{38}$$

$$\frac{R_{un}}{x_{max}^u} \geq R_{min}, \quad \forall n. \tag{39}$$

$$\widetilde{E}_{\mathrm{p}}(\mathcal{V}, \{o_n\}) + \sum_{n=1}^{N}\tau_n P_{un} \leq E_{max}. \tag{40}$$

$$\Lambda^{lb,r} \geq \frac{1}{o_n^2}, \forall n. \tag{41}$$

$$\mathrm{s.t.} \quad (25) \text{ and } (26). \tag{42}$$

4.1.2. UAV Waypoint and Time Duration Optimization

Optimize UAV waypoint $\mathcal{Q}$ and time duration T given $\mathcal{V}$, $\mathcal{A}$ as well as $\mathcal{X}$. The slack variables o_n are fixed at their latest feasible values in this subproblem. Equation (22) can be reduced to

$$(\text{P3.2}): \max_{\mathcal{Q},\mathcal{T},\{R_{un}\}} \sum_{i=1}^{I}\sum_{n=1}^{N} \tau_n x_{ian} R_{ian} + \sum_{n=1}^{N} \tau_n R_{un} \tag{43}$$

$$\text{s.t.} \quad (20) \text{ and } (21), \ (23)\text{–}(26). \tag{44}$$

Note that (43) is not a convex optimization problem because the objective function is non-concave and constraints (20) and (21) are non-convex. In particular, we can observe that $\tau_n R_{un}$ can be expressed as a difference of convex functions, i.e., $\tau_n R_{un} = \frac{1}{2}(\tau_n + R_{un})^2 - \frac{1}{2}(\tau_n^2 + R_{un}^2)$. To solve the non-concave objective function, using the first-order Taylor approximation on the $(\tau_n + R_{un})^2$ term in the r-th iteration given local points τ_n^r and R_{un}^r, we have $\tau_n R_{un} \geq (\tau_n^r + R_{un}^r)(\tau_n + R_{un}) - \frac{1}{2}(\tau_n^r + R_{un}^r)^2 - \frac{1}{2}(\tau_n^2 + R_{un}^2) \triangleq \Phi_{un}^{lb,r}$. Therefore, we replace $\tau_n R_{un}$ in the objective with $\Phi_{un}^{lb,r}$, and R_{iun} and R_{ubn} in the rate constraints with $\mathcal{R}_{iun}^{\text{lb,r}}$ and $\mathcal{R}_{ubn}^{\text{lb,r}}$, respectively. Problem (43) can be approximated as

$$(\text{P5}): \max_{\mathcal{Q},\mathcal{T},\{R_{un}\}} \sum_{i=1}^{I}\sum_{n=1}^{N} x_{ian}\tau_n R_{ian} + \sum_{n=1}^{N} \Phi_{un}^{lb,r} \tag{45}$$

$$\text{s.t.} \quad (23), (25) \text{ and } (26), (37) \text{ and } (38),$$
$$\sum_{n=1}^{N} \tau_n \left[P_0 + \frac{3P_0\|v_n^r\|^2}{U_{tip}^2} + \frac{1}{2} d_0 \rho s A \|v_n^r\|^3 + P_i o_n^r + P_{un} \right] \leq E_{max}. \tag{46}$$

It can be shown that (36) and (45) become convex optimization problems, which can be solved by existing tools such as CVX. Due to the lower bounds in (37), (38) and (41), the obtained solution of (36) can be used as a feasible solution of (29). Likewise, the solution obtained for (45) is a feasible solution for (43). The specific details are given in Algorithm 1.

Algorithm 1 DC-SCA-Based UAV Waypoint, UAV's Speed, and Time Duration Optimization for (22).

1: Initialize UAV waypoint $\{\mathcal{Q}^r\}$, time duration $\mathcal{T}^r$, UAV's speed $\mathcal{V}^r$, user connection scheduling $\{\mathcal{X}^r\}$, bandwidth allocation $\{\mathcal{A}^r\}$, and $\{R_{un}^r\}$ as well as $\{o_n^r\}$ using the current values from the outer iteration; set $r = l$.
2: Solve problem (36) to obtain $\mathcal{Q}^{r+1}$, $\mathcal{V}^{r+1}$ and $\{o_n^{r+1}\}$, $\{R_{un}^{r+1}\}$ with given local points $\mathcal{A}^r$, $\mathcal{X}^r$, and $\mathcal{T}^r$ using CVX;
3: Solve problem (45) to obtain $\mathcal{Q}^{r+1}$, $\mathcal{T}^{r+1}$, $\{R_{un}^{r+1}\}$ with given local points $\mathcal{A}^r$, $\mathcal{X}^r$, and the updated $\mathcal{V}^{r+1}$, $\{o_n^{r+1}\}$, $\mathcal{Q}^{r+1}$ using CVX.

4.2. User Connection Scheduling and Bandwidth Allocation Optimization

Through path discretization technology, we can get UAV waypoints $\mathcal{Q}$, UAV speed $\mathcal{V}$ and time duration $\mathcal{T}$. We assume that the discrete forms of x_{ia} and x_{iu} are x_{ian} and x_{iun} respectively. Let $\mathcal{X} = \{x_{ian}, x_{iun}, 1 \leq n \leq N\}$ and $\mathcal{A} = \{\alpha_{in}, 1 \leq n \leq N\}$. Problem (7) can be reduced to

$$(\text{P6}): \max_{\mathcal{A},\mathcal{X},\{R_{un}\}} \sum_{n=1}^{N}\sum_{i=1}^{I} \tau_n x_{ian} R_{ian} + \sum_{n=1}^{N} \tau_n R_{un} \tag{47}$$

$$\text{s.t.} R_{un} \leq \sum_{i=1}^{I} x_{iun} R_{iun}, \forall n, \tag{48}$$

$$\sum_{i=1}^{I} \alpha_{in} \leq 1, \alpha_{in} \in [0,1], \forall i, n, \tag{49}$$

$$x_{ian}, x_{iun} \in \{0,1\}, \forall i, n, \tag{50}$$

$$\sum_{i=1}^{I} x_{ian} \leq x_{max}^{a}, \forall n, \tag{51}$$

$$\sum_{i=1}^{I} x_{iun} \leq x_{max}^{u}, \forall n, \tag{52}$$

$$x_{ian} + x_{iun} \leq 1, \forall i, n, \tag{53}$$

$$\frac{\sum_{i=1}^{I} x_{ian} R_{ian}}{x_{max}^{a}} \geq R_{min}, \forall n, \tag{54}$$

$$\frac{R_{un}}{x_{max}^{u}} \geq R_{min}, \quad R_{un} \leq R_{ubn}, \quad \forall n. \tag{55}$$

We note that (47) is also a mixed-integer non-convex problem because of the objective function, binary variables (50), and constraints (48), (54) and (55). To solve this problem, we first need to deal with the binary constraint (50), and the constraints can be written equivalently as

$$\begin{aligned} &0 \leq x_{ian} \leq 1, 0 \leq x_{iun} \leq 1, \\ &\sum_{i=1}^{I} \sum_{n=1}^{N} (x_{ian} - (x_{ian})^2) \leq 0, \\ &\sum_{i=1}^{I} \sum_{n=1}^{N} (x_{iun} - (x_{iun})^2) \leq 0, \end{aligned} \tag{56}$$

where x_{ian} and x_{iun} are relaxed to continuous variables in $[0,1]$. In order to obtain the binary solution in (47), we introduce the method of penalization. Specifically, we introduce penalty terms in the objective function to discourage fractional scheduling values. Based on this approach, the objective function can be converted into $\sum_{i=1}^{I} \sum_{n=1}^{N} \tau_n x_{ian} R_{ian} + \sum_{n=1}^{N} \tau_n R_{un} - \chi \sum_{i=1}^{I} \sum_{n=1}^{N} (x_{ian} - (x_{ian})^2) - \xi \sum_{i=1}^{I} \sum_{n=1}^{N} (x_{iun} - (x_{iun})^2)$, where χ and ξ are positive penalty coefficients.

To handle the non-concave objective function and non-convex constraints, we relax the binary variables and introduce auxiliary variables Y, $\mathcal{Y} = \{y_{ian}, y_{iun}, \forall i, n\}$, and $\{R_{un}\}$. The penalized relaxation of problem (47) can then be written as

$$\begin{aligned} (\text{P6.1}): \max_{\mathcal{A}, \mathcal{X}, \mathcal{Y}, \{R_{un}\}, \text{Y}} \quad & \text{Y} - \chi \sum_{i=1}^{I} \sum_{n=1}^{N} (x_{ian} - (x_{ian})^2) \\ & - \xi \sum_{i=1}^{I} \sum_{n=1}^{N} (x_{iun} - (x_{iun})^2) \end{aligned} \tag{57}$$

$$\text{s.t.} \sum_{n=1}^{N} \sum_{i=1}^{I} \tau_n x_{ian} y_{ian} + \sum_{n=1}^{N} \tau_n R_{un} \geq \text{Y}. \tag{58}$$

$$R_{un} \leq \sum_{i=1}^{I} x_{iun} y_{iun}, \forall n, \tag{59}$$

$$\frac{\sum_{i=1}^{I} x_{ian} y_{ian}}{x_{max}^{a}} \geq R_{min}, \forall n, \tag{60}$$

$$\frac{R_{un}}{x_{max}^{u}} \geq R_{min}, \quad \forall n. \tag{61}$$

$$0 \leq x_{ian} \leq 1, \quad 0 \leq x_{iun} \leq 1, \quad y_{ian}, y_{iun}, R_{un} \geq 0, \quad \forall i, n. \tag{62}$$

$$y_{ian} \leq \alpha_{in} B \log_2 \left(1 + \frac{P_{in}\gamma_0}{\alpha_{in}(H_{bs}^2 + \|\mathbf{q}_A - \mathbf{q}_i^A\|^2)^{\alpha}}\right), \quad \forall i, n, \tag{63}$$

$$y_{iun} \leq \alpha_{in} B \log_2 \left(1 + \frac{P_{in}\gamma_0'}{\alpha_{in}(H_{uav}^2 + \|\mathbf{q}_n - \mathbf{q}_i^A\|^2)^{\alpha}}\right), \quad \forall i, n, \tag{64}$$

$$(21),\ (49),\ (51)\text{–}(53). \tag{65}$$

Constraints (63) and (64) are convex because their right-hand sides are concave functions of α_{in}.

However, the optimization problem in Equation (57) remains non-convex due to the non-concave objective function and non-convex constraints in (58)–(60). In particular, the bilinear term on the left-hand sides of (58) and (60) can be expressed as a DC function, i.e., $x_{ian}y_{ian} = \frac{1}{2}(x_{ian} + y_{ian})^2 - \frac{1}{2}((x_{ian})^2 + (y_{ian})^2)$. Therefore, with its first-order Taylor expansion around the local point (x_{ian}^r, y_{ian}^r) in the r-th iteration, the convex function $(x_{ian} + y_{ian})^2$ is approximated by $(x_{ian}^r + y_{ian}^r)^2 + 2(x_{ian}^r + y_{ian}^r)(x_{ian} + y_{ian} - x_{ian}^r - y_{ian}^r)$. As a result, we have

$$\begin{aligned} x_{ian}y_{ian} \geq\ & (x_{ian}^r + y_{ian}^r)(x_{ian} + y_{ian}) \\ & - \frac{1}{2}(x_{ian}^r + y_{ian}^r)^2 - \frac{1}{2}(x_{ian}^2 + y_{ian}^2) \\ \triangleq\ & \Psi_{ian}^{lb,r}. \end{aligned} \tag{66}$$

Similarly, it is observed that $x_{iun}y_{iun}$ in (59) can also be expressed as a difference of convex functions, i.e., $x_{iun}y_{iun} = \frac{1}{2}(x_{iun} + y_{iun})^2 - \frac{1}{2}(x_{iun}^2 + y_{iun}^2)$. Thus, with the local point (x_{iun}^r, y_{iun}^r), we have

$$\begin{aligned} x_{iun}y_{iun} \geq\ & (x_{iun}^r + y_{iun}^r)(x_{iun} + y_{iun}) \\ & - \frac{1}{2}(x_{iun}^r + y_{iun}^r)^2 - \frac{1}{2}(x_{iun}^2 + y_{iun}^2) \\ \triangleq\ & \widehat{\Gamma}_{iun}^{lb,r}. \end{aligned} \tag{67}$$

Besides, to tackle the non-concavity of the objective function, we can use the SCA optimization technique. Therefore, with the local points x_{ian}^r and x_{iun}^r, the objective function is lower bounded by $\mathrm{Y} - \chi\sum_{i=1}^{I}\sum_{n=1}^{N}(x_{ian} - x_{ian}^2) - \xi\sum_{i=1}^{I}\sum_{n=1}^{N}(x_{iun} - x_{iun}^2) \geq \mathrm{Y} - \chi\sum_{i=1}^{I}\sum_{n=1}^{N}\left(x_{ian} - (x_{ian}^r)^2 - 2x_{ian}^r(x_{ian} - x_{ian}^r)\right) - \xi\sum_{i=1}^{I}\sum_{n=1}^{N}\left(x_{iun} - (x_{iun}^r)^2 - 2x_{iun}^r(x_{iun} - x_{iun}^r)\right) \triangleq \mathrm{Y}^{\mathrm{lb},r}$. After the above changes, we obtain an approximation to (57)

$$(\mathrm{P7}): \max_{\mathcal{A},\mathcal{X},\mathcal{Y},\{R_{un}\},\mathrm{Y}} \mathrm{Y}^{lb,r} \tag{68}$$

$$\text{s.t.} \sum_{n=1}^{N}\sum_{i=1}^{I}\tau_n\Psi_{ian}^{lb,r} + \sum_{n=1}^{N}\tau_n R_{un} \geq \mathrm{Y}. \tag{69}$$

$$R_{un} \leq \sum_{i=1}^{I}\widehat{\Gamma}_{iun}^{lb,r}, \forall n, \tag{70}$$

$$\frac{\sum_{i=1}^{I}\Psi_{ian}^{lb,r}}{x_{max}^a} \geq R_{min}, \forall n = 1, \ldots, N, \tag{71}$$

$$\frac{R_{un}}{x_{max}^u} \geq R_{min}, \quad \forall n. \tag{72}$$

$$(63) \text{ and } (64), \quad y_{ian} \geq 0, \quad y_{iun} \geq 0, \quad R_{un} \geq 0, \quad \forall i, n. \tag{73}$$

$$(21), (49), (51)\text{–}(53), (62). \tag{74}$$

It is observed that (68) is a standard convex optimization problem, which can be efficiently solved by standard convex optimization techniques such as CVX. Algorithm 2 gives the specific algorithm details for solving problem (68).

Algorithm 2 Penalty-DC-Based User Connection Scheduling and Bandwidth Allocation Optimization for (47).

1: Initialize UAV trajectory $\{\mathcal{Q}^{r+1}\}$, time duration $\mathcal{T}^{r+1}$, speed $\mathcal{V}^{r+1}$, user connection scheduling $\{\mathcal{X}^r\}$, bandwidth allocation $\{\mathcal{A}^r\}$ and $\mathcal{Y}^r$ using the current values from the outer iteration; set $r = l$.
2: With given local points $\mathcal{Q}^{r+1}$, $\mathcal{T}^{r+1}$ and $\mathcal{V}^{r+1}$, obtain $\mathcal{A}^{r+1}, \mathcal{X}^{r+1}, \mathcal{Y}^{r+1}, \{R_{un}^{r+1}\}, \mathrm{Y}^{r+1}$ for (68) using CVX.

4.3. Overall Alternating Algorithm, Convergence and Complexity

According to the above discussion, we summarize the alternating optimization algorithm in Algorithm 3. The key is iterative optimization of the UAV's trajectory, UAV's speed, time duration, user connection scheduling, and bandwidth allocation. Specifically, the UAV's trajectory and speed form the first sub-block, yielding the convex optimization problem (36). The UAV's trajectory and time duration form the second sub-block, yielding the convex optimization problem (45), while (68) contains the remaining optimization variables. These three blocks are optimized alternately to obtain the final solution.

Let $\mathcal{Q}^l, \mathcal{V}^l, \mathcal{T}^l, \mathcal{X}^l, \mathcal{A}^l$ denote the optimization variables at the l-th iteration, and let R^l denote the computed throughput. The SCA and DC approximations provide locally tight lower bounds. With a feasible reference point, solving each convex subproblem does not decrease its corresponding surrogate objective. However, since the scheduling subproblem includes penalty terms, this does not directly guarantee that the original throughput increases after every scheduling update. Therefore, convergence is evaluated using the relative throughput change and scheduling integrality residual:

$$\delta^l = \frac{|R^l - R^{l-1}|}{\max\{1, |R^l|\}}, \quad l \geq 2. \tag{75}$$

The scheduling integrality residual ρ^l is the maximum absolute value of $x(1-x)$ over all scheduling variables. For the first iteration, $\delta^1 = +\infty$. The algorithm terminates when $\delta^l < \epsilon$ and $\rho^l < 10^{-3}$, or the maximum number of iterations is reached. After convergence, the near-binary schedule is projected to binary values, followed by bandwidth and rate optimization with the trajectory, time durations, and schedule fixed.

Algorithm 3 P-BCD-Based Optimization Algorithm for (7).

1: Initialize UAV trajectory $\{\mathcal{Q}^0\}$ as well as $\mathcal{V}^0, \mathcal{T}^0, \mathcal{X}^0, \mathcal{A}^0, \mathcal{Y}^0, \{o_n^0\}, \{R_{un}^0\}$; set $l = 0$.
2: **repeat**
3: With given local points $\mathcal{X}^l, \mathcal{A}^l$, obtain the updated values $\mathcal{Q}^{l+1}, \mathcal{V}^{l+1}, \mathcal{T}^{l+1}, \{R_{un}^{l+1}\}$ and $\{o_n^{l+1}\}$ with Algorithm 1;
4: With fixed local points $\mathcal{Q}^{l+1}, \mathcal{V}^{l+1}, \mathcal{T}^{l+1}$, obtain the updated values $\mathcal{X}^{l+1}, \mathcal{A}^{l+1}, \{R_{un}^{l+1}\}$ and $\mathcal{Y}^{l+1}$ with Algorithm 2;
5: Evaluate R^{l+1} and the stopping measures δ^{l+1}, ρ^{l+1}; update $l = l + 1$;
6: **until** ($\delta^l < \epsilon$ and $\rho^l < 10^{-3}$), or $l = 100$.

The complexity of Algorithm 3 is estimated as $O\big(L[2N^{3.5} + (IN)^{3.5}]\big)$, where L is the number of outer iterations, and $O(N^{3.5})$ and $O((IN)^{3.5})$ are the approximate costs of a trajectory subproblem and the scheduling/bandwidth subproblem, respectively. A final fixed-schedule bandwidth optimization requires one additional resource solve.

5. Results and Analysis

In this section, we conduct extensive simulations to evaluate the effectiveness of our algorithm and conduct comparative experiments with other benchmark algorithms.

5.1. Simulation Setup

We consider an OFDMA cellular network that includes two ground base stations (BSA and BSB) and a rotary-wing UAV that jointly provides communication services to ground users. To avoid signal occlusion, micro base stations in cities are usually placed at specific heights, so we assume that two homogeneous BSs have the same height H_{bs} = 25 m and coverage radius R = 300 m. Thus, in an area of 1 km × 1 km, the coordinates of BSA are $q_A = [300, 300, 25]^T$, $I = 10$ ground users are randomly distributed within its coverage, and the coordinates of BSB are $q_B = [720, 720, 25]^T$. Unless otherwise stated, we set the ratio of the maximum number of connectable ground users between BSA and the UAV to 7:3. The default settings are $N = 80$, $E_{max} = 20$ kJ, and $R_{min} = 0.2$ Mbps.

For the UAV, to avoid frequent ascent and descent, it is assumed that the flight altitude of the UAV is H_{uav} = 100 m, the maximum speed is set to be v_{max} = 30 m/s, $a_{max} = 5$ m/s^2, and, following the propulsion model in [41], the UAV energy-consumption-related parameters are as follows: d_0 = 0.6, U_{tip} = 120, A = 0.503, ρ = 1.225, s = 0.05, v_0 = 4.03, P_0 = 79.86, and P_i = 88.5. For the probabilistic LoS model, the related parameters are set as C = 10 and D = 0.6 [44]. Moreover, the regularized homogeneous LoS probability $\bar{P}_{\mathrm{LoS}}(t)$ is set as the value corresponding to the elevation angle of 45°, B = 10 MHz, P_{in} = 1 W, P_{un} = 1 W, κ = 0.2, $\widetilde{\alpha}$ = 2.2, $N_0 = -174$ dBm/Hz, and $\beta_0 = -50$ dB.

The parameters related to algorithm design are set as follows: d_{max} = 30 m, $\chi = \xi = 10^4$, and $\epsilon = 10^{-3}$. To start Algorithm 1, we set the initial UAV trajectory to a circle whose center is the geometric center of the ground users, similar to [23]. Unless otherwise stated, statistical results are averaged over five independent GU deployments generated with fixed random seeds. All simulations were performed using MATLAB R2025b and CVX 2.2 on a MacBook Air equipped with an Apple M5 processor and 16 GB of RAM.

5.2. Algorithm Convergence

Figure 2 shows the convergence of Algorithm 3 under different N values. For the same GU deployment, the algorithm converges after 22, 24, and 16 outer iterations for $N = 60$, 70, and 80, respectively. Among these settings, $N = 80$ achieves the highest throughput of 20,828.36 Mbit with the fewest iterations. Based on this comparison, we set $N = 80$ in the remaining simulations. The plotted throughput is evaluated before the final fixed-schedule bandwidth refinement.

5.3. Penalty Sensitivity and Computational Cost

Table 2 compares different penalty coefficients on a fixed GU deployment under the default settings. All four settings converge in 16 iterations, with integrality residuals below 5.2×10^{-9} and throughput differences below 0.02%. Therefore, we set $\chi = \xi = 10^4$ in the remaining simulations. The table reports throughput after the final fixed-schedule bandwidth refinement, whereas residuals and runtimes are measured before refinement.

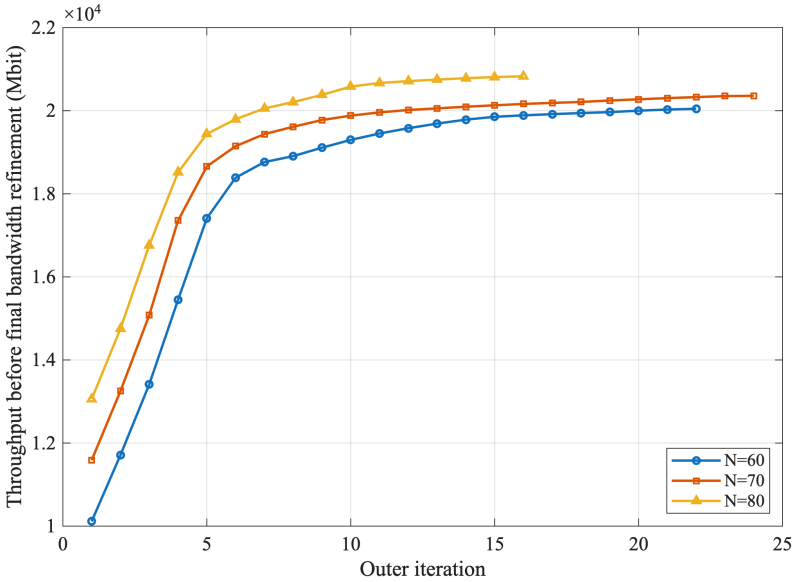


Figure 2. Convergence of Algorithm 3.

Table 2. Penalty sensitivity under the default settings.

$\chi = \xi$	Final Throughput (Mbit)	Integrality Residual	Outer Iterations	Runtime (s)
10^2	21,204.204	9.54×10^{-10}	16	2567.25
10^3	21,204.110	7.75×10^{-10}	16	2485.49
10^4	21,202.885	3.12×10^{-9}	16	2390.91
10^5	21,200.082	5.15×10^{-9}	16	2235.89

Table 3 presents representative module runtimes. One measured outer iteration takes 94.69 s, with the scheduling and bandwidth update accounting for approximately 70.1% of this time.

Table 3. Representative module runtimes under the default settings.

Operation	Runtime (s)
Trajectory, speed, and time update	28.33
Scheduling and bandwidth update	66.36
Total for one measured outer iteration	94.69
Final fixed-schedule bandwidth refinement	15.96

5.4. UAV Trajectory and UAV Speed as Well as User Connection Scheduling

Figure 3 shows the trajectories of the optimized UAV under different UAV energy budgets E_{max} and QoS requirements R_{min}. It can be observed that the UAV follows similar routes under the three settings, with local differences highlighted in the inset. Increasing the energy budget or the QoS requirement mainly changes local waypoint positions rather than substantially expanding the flight region. Figure 4 shows that when the energy budget $E_{max} = 20$ kJ is the same, the UAV speed varies differently under $R_{min} = 0.2$ and 5 Mbps, while the service durations remain similar at 154.03 and 152.26 s, respectively. In Figure 4, we can also observe that as the energy budget increases, the UAV service time also increases. Specifically, increasing E_{max} from 20 to 50 kJ at $R_{min} = 0.2$ Mbps extends the service time to 359.94 s, with the UAV spending longer periods at lower speeds. Together with the similar trajectories in Figure 3, this indicates that the additional energy is mainly used to extend communication service in these cases.

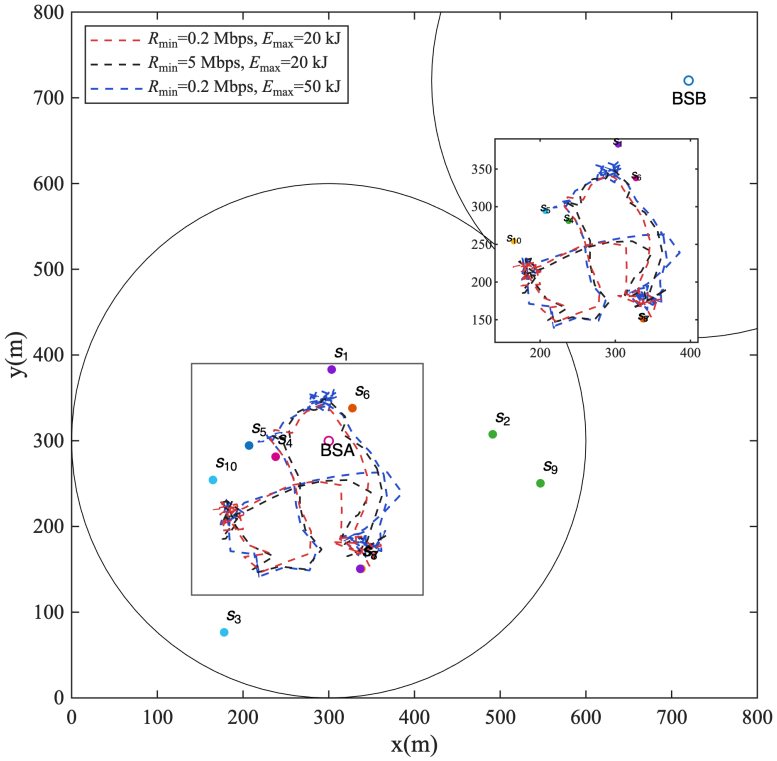


Figure 3. Optimized UAV trajectories under different energy budgets and QoS requirements. The dashed lines denote the parameter settings shown in the legend; markers along the trajectories indicate UAV waypoints, filled colored circles denote GUs, open circles denote the two BSs, and black circular outlines indicate their coverage boundaries.

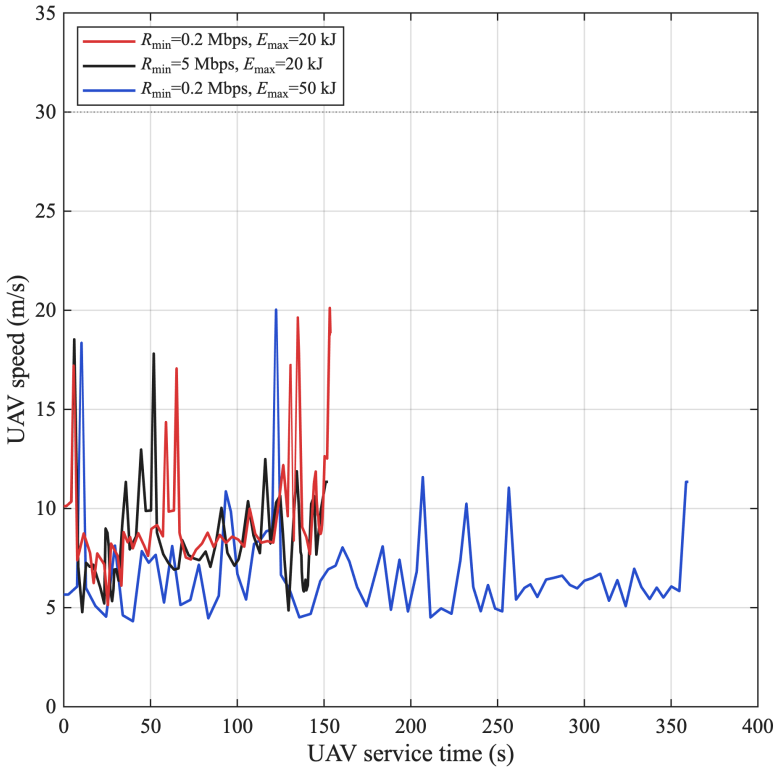


Figure 4. Optimized UAV speed versus service time under the same settings as Figure 3.

Table 4 reports the average percentage of segments in which the QoS constraints are active over five GU deployments. A constraint is considered active when the corresponding

mode-average rate is within 10^{-4} Mbps of R_{min}. For the tested thresholds of 0.2, 0.5, and 1.5 Mbps, neither QoS constraint is active. At 5 and 10 Mbps, the relay QoS constraint is active in 11% and 42.5% of segments, respectively, while the cellular QoS constraint remains nonbinding. Thus, higher thresholds make the relay QoS constraint more frequently binding in these experiments.

Table 4. Mean QoS constraint active fractions over five GU deployments.

R_{min} (Mbps)	Cellular QoS Active Fraction (%)	UAV Relay QoS Active Fraction (%)
0.2	0	0
0.5	0	0
1.5	0	0
5	0	11
10	0	42.5

Figure 5 compares user connection scheduling under connection budgets 7:3 and 4:2 on the same GU deployment, with $E_{max} = 20$ kJ and $R_{min} = 0.2$ Mbps. Users switch between cellular and UAV relay modes along the flight path. Under budgets 7:3, all ten users are served in each segment, whereas under budgets 4:2, six users are served simultaneously and the remaining users are inactive. The changing relay assignments illustrate how scheduling adapts to the UAV's position.

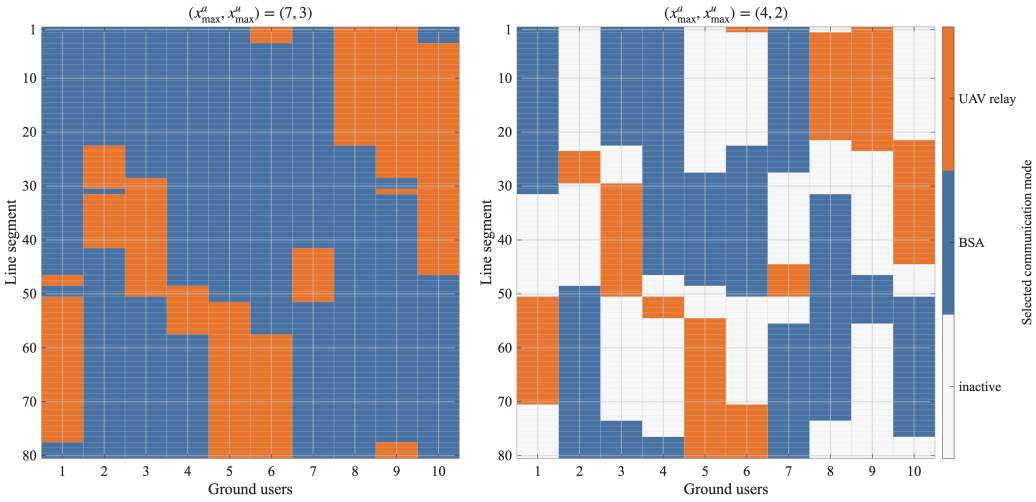


Figure 5. Optimized user connection scheduling under budgets 7:3 and 4:2 for the same GU deployment. Orange, blue, and white denote the UAV relay mode, BSA cellular mode, and inactive users, respectively.

5.5. Performance Comparison

In this part, we compare different UAV trajectories, user connection scheduling strategies, and bandwidth allocation strategies to evaluate their effects on regional communication capacity. We use the following benchmark schemes to verify the effectiveness of the proposed algorithm:

- No-UAV benchmark: The UAV is unavailable in the considered scenario.
- Distance-based k-means clustering benchmark: The k-means clustering algorithm is used to divide 10 GUs into four mutually exclusive clusters, and relay users are selected according to the distance between the current UAV position and the geometric centers of the clusters, subject to the connection budgets, as described in [45].
- Circular UAV trajectory benchmark: The UAV flies along a circular trajectory centered at the geometric center of GUs with a fixed radius. User connection scheduling and

bandwidth allocation are optimized with the trajectory and segment durations fixed, similar to [23].

- Random bandwidth allocation benchmark: For each line segment, the bandwidth is assigned randomly and kept fixed while optimizing the UAV trajectory, service time, and user connection scheduling, as described in [46].

We further consider the following variants to examine the effects of individual optimization components:

- Fixed hovering benchmark: The UAV hovers at the midpoint between BSA and BSB. Binary user assignments are enumerated, and bandwidth allocation is optimized for each assignment.
- Resources-only optimization: The initial UAV trajectory and segment durations are fixed, while user connection scheduling and bandwidth allocation are optimized.
- Trajectory-only optimization: User connection scheduling and bandwidth allocation are fixed at their initial feasible values, while the UAV trajectory and service time are optimized.
- Greedy scheduling benchmark: Users are selected using a greedy scheduling rule, while the UAV trajectory, service time, and bandwidth allocation are optimized.
- SCA without binary penalties: Scheduling variables are relaxed to continuous values and optimized using SCA without binary penalty terms, followed by binary schedule recovery.

Figure 6 compares the regional communication capacity achieved by different benchmark schemes and optimization variants. The proposed algorithm achieves the highest mean capacity of 24,207.41 Mbit, while the No-UAV benchmark achieves 12,877.78 Mbit over a fixed service time of 80 s. Since the UAV schemes use their respective service durations, this difference reflects both communication performance and service time. Among the three UAV benchmark schemes, distance-based k-means clustering achieves the highest mean capacity of 23,089.91 Mbit, followed by the circular UAV trajectory benchmark at 22,320.40 Mbit and the random bandwidth allocation benchmark at 20,931.94 Mbit. Compared with these schemes, the proposed algorithm jointly optimizes UAV trajectory, service time, user connection scheduling, and bandwidth allocation, achieving higher mean capacity in the tested deployments. The ablation results further illustrate the effects of individual optimization components. Fixed hovering and resources-only optimization yield lower capacities, showing the benefit of optimizing UAV mobility and service time. Trajectory-only optimization, greedy scheduling, and SCA without binary penalties achieve capacities closer to, but below, the proposed algorithm. Results show means and 95% confidence intervals over five GU deployments, except for SCA without binary penalties, which uses four successful deployments.

Figure 7 shows that as the UAV energy budget E_{max} increases from 20 to 60 kJ, the mean capacity increases from 24,207.41 to 65,031.13 Mbit, while the mean service time increases from 151.29 to 407.79 s. This is because a larger energy budget allows the UAV to provide communication services for a longer duration, thereby increasing the total transmitted data. Results are averaged over the same five GU deployments, with error bars indicating 95% confidence intervals. Figure 8 shows the regional communication capacity under different QoS requirements, using the same five GU deployments. As R_{min} increases from 0.2 to 10 Mbps, the mean capacity decreases slightly from 24,207.41 to 24,031.28 Mbit, a reduction of approximately 0.73%. Higher QoS requirements restrict resource allocation, but the observed capacity change is small relative to the variation across deployments. Both connection budgets remain fully utilized throughout the tested range; therefore, the decrease is not caused by serving fewer users.

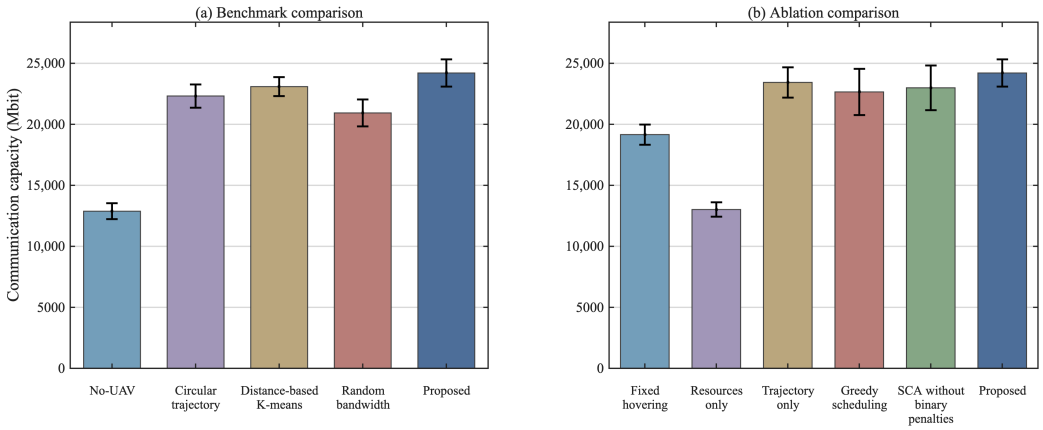


Figure 6. Regional communication capacity comparison: (**a**) benchmark schemes and (**b**) optimization variants.

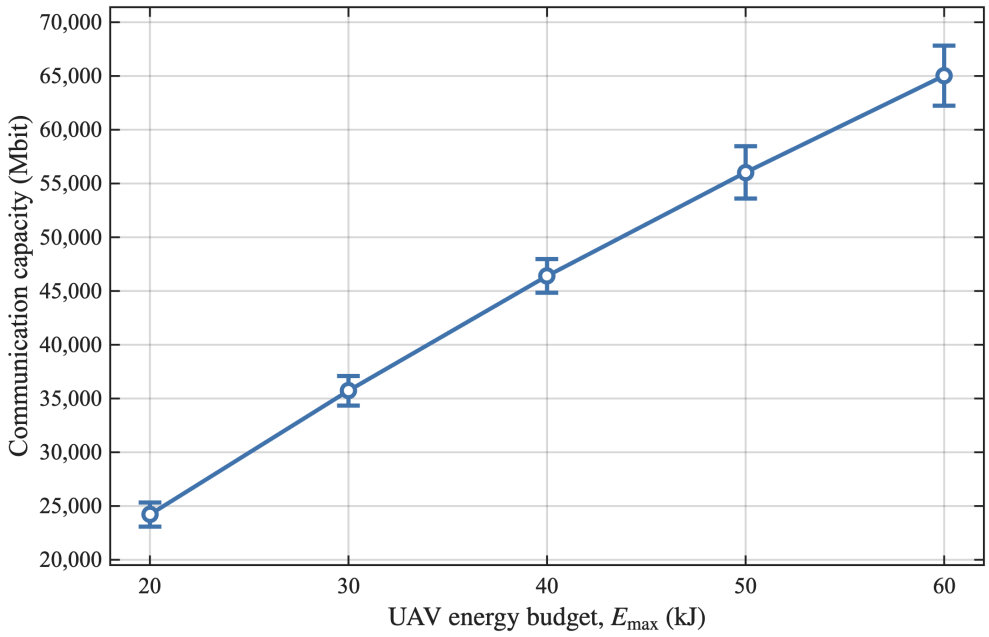


Figure 7. Communication capacity versus UAV energy budget.

5.6. Limitations

The current model considers a single UAV at a fixed altitude, static GUs, and known channel conditions, with fully compensated Doppler shifts. It uses ideal two-hop rate constraints without explicitly modeling access–backhaul interference, buffering delay, or communication-mode switching overhead [47,48]. In practice, interference, channel fluctuations, and residual Doppler shifts may reduce achievable rates, while mode switching and buffering may introduce additional delays. Although the backhaul rate limits relay throughput in the model, this rate constraint does not provide an end-to-end latency guarantee [49]. Therefore, the reported performance should be interpreted under the adopted assumptions.

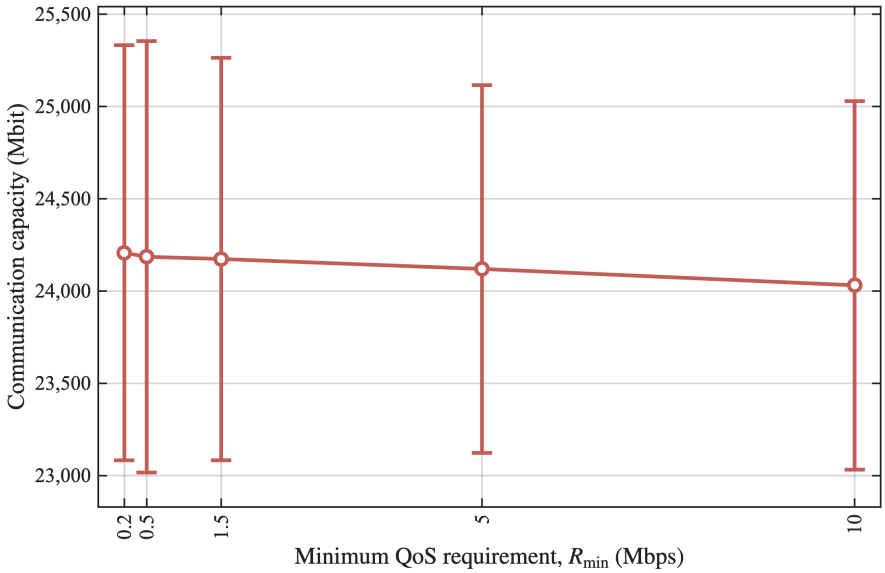


Figure 8. Communication capacity versus QoS requirement.

6. Conclusions and Future Work

In this paper, we investigated an energy-efficient UAV relay-assisted communication system operating in dual communication mode, aiming to mitigate congestion in high-demand cellular networks. To maximize the long-term regional communication capacity, we formulated a joint optimization problem incorporating user connection scheduling, bandwidth allocation, and UAV flight trajectory under UAV energy budgets and two-hop QoS constraints. To tackle the resulting mixed-integer nonlinear non-convex problem (MINLP), we developed an algorithmic framework combining path discretization with block coordinate descent (BCD), successive convex approximation (SCA), difference-of-convex programming, and a penalty-DC method. Simulation results show that the proposed algorithm achieves higher mean regional communication capacity than the tested UAV benchmark schemes. Furthermore, our findings highlight the trade-offs between UAV trajectories, onboard energy limitations, and user QoS requirements, while demonstrating the effectiveness of the penalty-DC method in obtaining near-binary scheduling solutions.

In future work, this framework can be extended to multi-UAV relay networks by incorporating inter-drone collision avoidance and channel interference mitigation. Additionally, the integration of Large Language Models (LLMs) and advanced Artificial Intelligence (AI) techniques, taking into account user mobility, imperfect channel information, and communication-mode switching overhead, will further improve the applicability of the proposed framework to practical cellular networks [50].

Author Contributions: Conceptualization, Y.H. and J.K.; methodology, Y.H.; software, Y.H.; validation, Y.H. and J.K.; formal analysis, Y.H.; investigation, Y.H.; resources, J.K.; data curation, Y.H.; writing—original draft preparation, Y.H.; writing—review and editing, J.K.; visualization, Y.H.; supervision, J.K.; project administration, J.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

UAV	Unmanned aerial vehicle
GU	Ground user
BS	Base station
OFDMA	Orthogonal frequency-division multiple access
LoS	Line of sight
NLoS	Non-line of sight
QoS	Quality of service
QoE	Quality of experience
MEC	Mobile edge computing
MD	Mobile device
FD	Full duplex
IoT	Internet of Things
BCD	Block coordinate descent
P-BCD	Path-discretization-based block coordinate descent
SCA	Successive convex approximation
DC	Difference of convex
AO	Alternating optimization
MINLP	Mixed-integer nonlinear programming

Appendix A

Let Δe_n denote the finite-sum approximation error on segment n, with the communication variables held fixed as specified above. By the Lipschitz property,

$$\begin{aligned} |\Delta e_n| &= \left| \int_0^{\tau_n} [R_u(q_{n-1} + tv_n) - R_{un}(q_n)]dt \right| \\ &\leq \int_0^{\tau_n} |R_u(q_{n-1} + tv_n) - R_{un}(q_n)|dt \\ &\leq D_{qn} \int_0^{\tau_n} \|q_{n-1} + tv_n - q_n\|dt \\ &= D_{qn}\|v_n\| \int_0^{\tau_n} (\tau_n - t)dt \\ &= \frac{1}{2} D_{qn}\|v_n\|\tau_n^2 = \frac{1}{2} D_{qn} d_n \tau_n, \end{aligned} \tag{A1}$$

where $q_n = q_{n-1} + \tau_n v_n$ and $d_n = \|v_n\|\tau_n$. Applying the triangle inequality across all segments gives

$$\begin{aligned} e &= \left| \int_0^T R_u(q(t))dt - \sum_{n=1}^N \tau_n R_{un}(q_n) \right| \\ &= \left| \sum_{n=1}^N \Delta e_n \right| \leq \sum_{n=1}^N |\Delta e_n| \\ &\leq \frac{1}{2} \sum_{n=1}^N D_{qn} d_n \tau_n \leq \frac{1}{2} D_q d_{max} T. \end{aligned} \tag{A2}$$

The last inequality follows from $D_q = \max_n D_{qn}$, $d_n \leq d_{max}$, and $T = \sum_n \tau_n$, completing the proof.

References

1. The Mobile Economy 2023. Technical Report, GSMA. 2023. Available online: https://www.gsma.com/mobileeconomy/ (accessed on 28 October 2026).
2. Aloqaily, M.; Ridhawi, I.A.; Salameh, H.B.; Jararweh, Y. Data and Service Management in Densely Crowded Environments: Challenges, Opportunities, and Recent Developments. *IEEE Commun. Mag.* **2019**, *57*, 81–87. [CrossRef]

3. Zeng, Y.; Zhang, R.; Lim, T.J. Wireless communications with unmanned aerial vehicles: Opportunities and challenges. *IEEE Commun. Mag.* **2016**, *54*, 36–42. [CrossRef]
4. Mozaffari, M.; Taleb Zadeh Kasgari, A.; Saad, W.; Bennis, M.; Debbah, M. Beyond 5G With UAVs: Foundations of a 3D Wireless Cellular Network. *IEEE Trans. Wirel. Commun.* **2019**, *18*, 357–372. [CrossRef]
5. Xu, C.; Zhan, C.; Liao, J.; Gong, J. Computation Throughput Maximization for UAV-Enabled MEC with Binary Computation Offloading. In Proceedings of the ICC 2022—IEEE International Conference on Communications, Seoul, Republic of Korea, 16–20 May 2022; pp. 4348–4353. [CrossRef]
6. Xu, Y.; Zhang, T.; Liu, Y.; Yang, D.; Xiao, L.; Tao, M. UAV-Assisted MEC Networks With Aerial and Ground Cooperation. *IEEE Trans. Wirel. Commun.* **2021**, *20*, 7712–7727. [CrossRef]
7. Lu, W.; Ding, Y.; Gao, Y.; Chen, Y.; Zhao, N.; Ding, Z.; Nallanathan, A. Secure NOMA-Based UAV-MEC Network Towards a Flying Eavesdropper. *IEEE Trans. Commun.* **2022**, *70*, 3364–3376. [CrossRef]
8. Zeng, B.; Zhan, C.; Xu, C.; Liao, J. Caching and 3D Deployment Strategy for Scalable Videos in Cache-Enabled Multi-UAV Networks. *IEEE Trans. Veh. Technol.* **2023**, *72*, 14875–14888. [CrossRef]
9. Zhang, T.; Wang, Y.; Yi, W.; Liu, Y.; Nallanathan, A. Joint Optimization of Caching Placement and Trajectory for UAV-D2D Networks. *IEEE Trans. Commun.* **2022**, *70*, 5514–5527. [CrossRef]
10. Zeng, Y.; Wu, Q.; Zhang, R. Accessing From the Sky: A Tutorial on UAV Communications for 5G and Beyond. *Proc. IEEE* **2019**, *107*, 2327–2375. [CrossRef]
11. Zhan, C.; Zeng, Y. Completion Time Minimization for Multi-UAV-Enabled Data Collection. *IEEE Trans. Wirel. Commun.* **2019**, *18*, 4859–4872. [CrossRef]
12. Ebrahimi, D.; Sharafeddine, S.; Ho, P.H.; Assi, C. UAV-Aided Projection-Based Compressive Data Gathering in Wireless Sensor Networks. *IEEE Internet Things J.* **2019**, *6*, 1893–1905. [CrossRef]
13. Samir, M.; Sharafeddine, S.; Assi, C.M.; Nguyen, T.M.; Ghrayeb, A. UAV Trajectory Planning for Data Collection from Time-Constrained IoT Devices. *IEEE Trans. Wirel. Commun.* **2020**, *19*, 34–46. [CrossRef]
14. Ding, R.; Chen, J.; Wu, W.; Liu, J.; Gao, F.; Shen, X. Packet routing in dynamic multi-hop UAV relay network: A multi-agent learning approach. *IEEE Trans. Veh. Technol.* **2022**, *71*, 10059–10072. [CrossRef]
15. Zeng, S.; Zhang, H.; Di, B.; Song, L. Trajectory Optimization and Resource Allocation for OFDMA UAV Relay Networks. *IEEE Trans. Wirel. Commun.* **2021**, *20*, 6634–6647. [CrossRef]
16. Zhang, S.; Zhang, H.; He, Q.; Bian, K.; Song, L. Joint Trajectory and Power Optimization for UAV Relay Networks. *IEEE Commun. Lett.* **2018**, *22*, 161–164. [CrossRef]
17. Kang, J.M.; Chun, C.J. Joint Trajectory Design, Tx Power Allocation, and Rx Power Splitting for UAV-Enabled Multicasting SWIPT Systems. *IEEE Syst. J.* **2020**, *14*, 3740–3743. [CrossRef]
18. Cui, J.; Shakhatreh, H.; Hu, B.; Chen, S.; Wang, C. Power-Efficient Deployment of a UAV for Emergency Indoor Wireless Coverage. *IEEE Access* **2018**, *6*, 73200–73209. [CrossRef]
19. Jin, J.; Yu, K.; Kua, J.; Zhang, N.; Pang, Z.; Han, Q.L. Cloud-Fog Automation: Vision, Enabling Technologies, and Future Research Directions. *IEEE Trans. Ind. Informat.* **2024**, *20*, 1039–1054. [CrossRef]
20. Jin, J.; Pang, Z.; Kua, J.; Zhu, Q.; Johansson, K.H.; Marchenko, N.; Cavalcanti, D. Cloud-Fog Automation: The New Paradigm Toward Autonomous Industrial Cyber-Physical Systems. *IEEE J. Sel. Areas Commun.* **2025**, *43*, 2917–2937. [CrossRef]
21. Diao, X.; Yang, W.; Yang, L.; Cai, Y. UAV-Relaying-Assisted Multi-Access Edge Computing With Multi-Antenna Base Station: Offloading and Scheduling Optimization. *IEEE Trans. Veh. Technol.* **2021**, *70*, 9495–9509. [CrossRef]
22. Parvaresh, N.; Kulhandjian, M.; Kulhandjian, H.; D'amours, C.; Kantarci, B. A tutorial on AI-powered 3D deployment of drone base stations: State of the art, applications and challenges. *Veh. Commun.* **2022**, *36*, 100474. [CrossRef]
23. Wu, Q.; Zhang, R. Common Throughput Maximization in UAV-Enabled OFDMA Systems With Delay Consideration. *IEEE Trans. Commun.* **2018**, *66*, 6614–6627. [CrossRef]
24. Sharafeddine, S.; Islambouli, R. On-demand deployment of multiple aerial base stations for traffic offloading and network recovery. *Comput. Netw.* **2019**, *156*, 52–61. [CrossRef]
25. Jia, D.; Kua, J.; Liu, X. A Lightweight LSTM Model for Flight Trajectory Prediction in Autonomous UAVs. *Future Internet* **2026**, *18*, 4. [CrossRef]
26. Jia, D.; Kua, J.; Liu, X. Performance Analysis of Network-Aware Micro-UAVs in Low-Altitude Applications. In Proceedings of the 2025 IEEE 50th Conference on Local Computer Networks (LCN), Sydney, Australia, 14–16 October 2025; pp. 1–7. [CrossRef]
27. Kua, J.; Armitage, G.; Branch, P. A Survey of Rate Adaptation Techniques for Dynamic Adaptive Streaming Over HTTP. *IEEE Commun. Surv. Tuts.* **2017**, *19*, 1842–1866. [CrossRef]
28. Xu, X.; Zeng, Y.; Guan, Y.L.; Zhang, R. Overcoming Endurance Issue: UAV-Enabled Communications with Proactive Caching. *IEEE J. Sel. Areas Commun.* **2018**, *36*, 1231–1244. [CrossRef]
29. Ladegourdie, M.; Kua, J. Performance Analysis of OPC UA for Industrial Interoperability towards Industry 4.0. *IoT* **2022**, *3*, 507–525. [CrossRef]

30. Zhu, L.; Zhang, J.; Xiao, Z.; Cao, X.; Xia, X.G.; Schober, R. Millimeter-Wave Full-Duplex UAV Relay: Joint Positioning, Beamforming, and Power Control. *IEEE J. Sel. Areas Commun.* **2020**, *38*, 2057–2073. [CrossRef]
31. Li, B.; Zhao, S.; Zhang, R.; Yang, L. Full-Duplex UAV Relaying for Multiple User Pairs. *IEEE Internet Things J.* **2021**, *8*, 4657–4667. [CrossRef]
32. Cheng, F.; Gui, G.; Zhao, N.; Chen, Y.; Tang, J.; Sari, H. UAV-Relaying-Assisted Secure Transmission With Caching. *IEEE Trans. Commun.* **2019**, *67*, 3140–3153. [CrossRef]
33. Yeom, J.S.; Kim, Y.B.; Jung, B.C. UAV-Assisted Cooperative Downlink NOMA with Virtual Full-Duplex Operation. In Proceedings of the 2019 Eleventh International Conference on Ubiquitous and Future Networks (ICUFN), Zagreb, Croatia, 2–5 July 2019; pp. 4–6. [CrossRef]
34. Zhong, X.; Guo, Y.; Li, N.; Chen, Y.; Li, S. Deployment Optimization of UAV Relay for Malfunctioning Base Station: Model-Free Approaches. *IEEE Trans. Veh. Technol.* **2019**, *68*, 11971–11984. [CrossRef]
35. Zhan, C.; Zeng, Y. Energy-Efficient Data Uploading for Cellular-Connected UAV Systems. *IEEE Trans. Wirel. Commun.* **2020**, *19*, 7279–7292. [CrossRef]
36. You, C.; Zhang, R. 3D Trajectory Optimization in Rician Fading for UAV-Enabled Data Harvesting. *IEEE Trans. Wirel. Commun.* **2019**, *18*, 3192–3207. [CrossRef]
37. Zeng, Y.; Xu, J.; Zhang, R. Energy minimization for wireless communication with rotary-wing UAV. *IEEE Trans. Wirel. Commun.* **2019**, *18*, 2329–2345. [CrossRef]
38. Lin, X.; Yajnanarayana, V.; Muruganathan, S.D.; Gao, S.; Asplund, H.; Maattanen, H.L.; Bergstrom, M.; Euler, S.; Wang, Y.P.E. The Sky Is Not the Limit: LTE for Unmanned Aerial Vehicles. *IEEE Commun. Mag.* **2018**, *56*, 204–210. [CrossRef]
39. Van Der Bergh, B.; Chiumento, A.; Pollin, S. LTE in the sky: Trading off propagation benefits with interference costs for aerial nodes. *IEEE Commun. Mag.* **2016**, *54*, 44–50. [CrossRef]
40. Matolak, D.W.; Sun, R. Air–Ground Channel Characterization for Unmanned Aircraft Systems—Part III: The Suburban and Near-Urban Environments. *IEEE Trans. Veh. Technol.* **2017**, *66*, 6607–6618. [CrossRef]
41. Zhan, C.; Huang, R. Energy Efficient Adaptive Video Streaming with Rotary-Wing UAV. *IEEE Trans. Veh. Technol.* **2020**, *69*, 8040–8044. [CrossRef]
42. Zhan, C.; Zeng, Y. Energy Minimization for Cellular-Connected UAV: From Optimization to Deep Reinforcement Learning. *IEEE Trans. Wirel. Commun.* **2022**, *21*, 5541–5555. [CrossRef]
43. Guo, Y.; You, C.; Yin, C.; Zhang, R. UAV Trajectory and Communication Co-Design: Flexible Path Discretization and Path Compression. *IEEE J. Sel. Areas Commun.* **2021**, *39*, 3506–3523. [CrossRef]
44. Zhan, C.; Zeng, Y. Aerial–Ground Cost Tradeoff for Multi-UAV-Enabled Data Collection in Wireless Sensor Networks. *IEEE Trans. Commun.* **2020**, *68*, 1937–1950. [CrossRef]
45. Tran, T.N.; Nguyen, T.L.; Hoang, V.T.; Voznak, M. Sensor Clustering Using a K-Means Algorithm in Combination with Optimized Unmanned Aerial Vehicle Trajectory in Wireless Sensor Networks. *Sensors* **2023**, *23*, 2345. [CrossRef]
46. Mahmood, A.; Vu, T.X.; Sharma, S.K.; Chatzinotas, S.; Ottersten, B. Multi-objective optimization for 3D placement and resource allocation in ofdma-based multi-uav networks. *arXiv* **2023**, arXiv:2304.12798.
47. Hao, Y.; Kua, J. Joint Scheduling and 3D UAV Deployment for Peak Age of Information Minimization in Last-Mile Logistics Networks. *Drones* **2026**, *10*, 741. [CrossRef]
48. Mahmud, R.; Jin, J.; Kua, J.; Afrin, M.; Mistry, S.; Krishna, A. Trusted Microservice Orchestration for Secure Edge Computing in Industrial Cyber-Physical Systems. *IEEE Netw.* **2025**, *39*, 70–78. [CrossRef]
49. Wang, S.; Kua, J.; Jin, J.; Wong, Y.W.; Jayaraman, P.P.; Pang, Z. Assisting mission-critical traffic flows with Active Queue Management in Industrial Internet of Things. *J. Ind. Inf. Integr.* **2026**, *53*, 101170. [CrossRef]
50. Javaid, S.; Fahim, H.; He, B.; Saeed, N. Large language models for UAVs: Current state and pathways to the future. *IEEE Open J. Veh. Technol.* **2024**, *5*, 1166–1192. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.